Fiscal Year:	FY 2019	**Task Last Updated:**	FY 08/06/2018
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Space Flight Environment Induces Remodeling of Vascular Network and Glia-Vascular Communication in Mouse Retina		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	xmao@llu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8373
Organization Name:	Loma Linda University		
PI Address 1:	Radiation Medicine		
PI Address 2:	11175 Campus Street, Rm A1010, Chan Shun Pavilion		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Rodent Research-9 NOTE: End date changed to 10/31/2021 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: End date changed to 3/31/2020 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 4/30/2019 per F. Hernandez/ARC (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University) Sun, Shu-Wei Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX15AB41G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The eye is a unique organ because it is relatively unprotected and constantly exposed to atmospheric oxygen, background radiation, environmental chemicals, and physical abrasion. A recent report shows that more than 30% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. Our Preliminary studies from mice that had been subjected to space flights showed that environmental conditions during space travel lead to oxidative stress and induce adverse microvessel remodeling in the retina. However, our knowledge about the pathophysiological process from adaptive response to irreversible oxidative damage in the retina during and after space flight exposure and the underlying cellular mechanism(s) of these disturbances is very limited and inconsistent. The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. The hypothesis will be tested in the specific aims with the space flight experiments on the International Space Station (ISS) and compared to that of ground-based groups using matured female mice. For flight studies, there will be two experiments over two flights, one for in-flight adaptation study on the ISS and one for studying re-adaptation after live animal return to Earth following long-term ISS mission. Selected groups of mice will be treated intraperitoneally with an antioxidant a week before launch and weekly during space flight. Animals will be sacrificed in orbit at 14 and 30 days after launch or 30 days and up to 6 months after return alive on Earth. Together, our unique, integrative, and quantitative activities with advanced MRI (magnetic resonance imaging) imaging techniques and retinal functional testing using visual electroretinography will provide insight into the molecular mechanisms and pathways of space flight condition-induced mitochondrial oxidative damage on retinal BRB integrity, vascular remodeling, and retinal function. The data will provide unique criteria for risks of functional detriments associated with space flight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Task Progress:	The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against spaceflight environment-induced changes in retinal neurovascular coupling and retinal function. SpaceX-12 was launched at the Kennedy Space Center (KSC) on a 30-day mission on August 14, 2017. Nine week old male C57BL/6 mice (n=20) were flown on SpaceX-12 in NASA's animal enclosure modules (AEMs). The animals then were returned to Loma Linda University (LLU) where they were subject to sacrifice followed by dissections. Experiment with ground-based control groups have been completed in May, 2018. Fixed ocular samples are under sectioning for histochemical analysis for identifying biomarkers for oxidative stress, apoptosis, and blood-retina-barrier (BRB) function. Frozen retina samples are prepared for genomics and metabolomics analysis. Our preliminary data showed that space flight conditions, especially gravitational changes, may induce acute and short-term response in intraocular pressure (IOP), most likely due to a compensatory mechanism. Global ocular structures were also measured using MicroCT. The cross section areas of retina pigment epithelium (RPE), retina, and choroid layer were significantly or trend lower than habitat and vivarium controls. Further investigation is needed to determine the consequences of the acute IOP changes on ocular structure and function. The data were presented at ISS R&D (Research & Development) and NASA Human Research Program Workshop, 2018. Text from the meeting abstract: "A recent report shows that more than 30% of the astronauts returning from space shuttle or International Space Station (ISS) missions have reported a subjective change in their eyesight. A dozen or more had documented anatomical eye changes post mission. These visual functional and anatomical disturbances have been hypothesized to be related to increases in intraocular pressure (IOP). The purpose of present study is to evaluate acute IOP and global ocular structure changes following spaceflight compared to pre-flight and vivarium controls. SpaceX-12 was launched at the Kennedy Space Center (KSC) on a 34-day mission on August 14, 2017. Nine week old male C57BL/6 mice (n=20) were flown on SpaceX-12 in NASA's animal enclosure modules (AEMs). Repeated IOP measurements were performed before the flight launch. A handheld tonometer was used for conscious mouse IOP measurement without a local anesthetic. Within 36 hours of splash-down, the spaceflight group were re-measured before they were [sacrificed] for further analysis. Age-matched vivarium mice which were housed in regular cages with regular food and water were only measured once at time of [sacrifice]. Cross sections of ocular layer were also measured using MicroCT on fixed eye. IOP was significantly lower in post-flight measurements compared to that of pre-flight (16.3-17.7 mmHg vs 17.6-19.5 mmHg mmHg) ($p<0.05$); age-matched controls had the lowest IOP measurement (14.4-16.2 mmHg). Eye side differences in IOP were also detected. Right eyes had lower values than that of right eyes in all groups. The cross section areas of retina pigment epithelium (RPE), retina and choroid layer were significantly or trend lower than vivarium controls. The rebound tonometer was easy to use and accurately measured IOP in mice. Our study indicates that space flight conditions, especially gravitational changes, may induce acute and short-term response in IOP, most likely due to a compensatory mechanism. Further investigation is needed to determine the consequences of the acute IOP changes on ocular structure and function. Additional studies are also needed on long-term regulation of ocular hemodynamics on IOP following changes in gravitational forces."
Bibliography Type:	Description: (Last Updated: 07/10/2026)

Abstracts for Journals and Proceedings	Mao XW, Nishiyama NC, Pecaut MJ. "Acute effect of spaceflight on ocular structure and intraocular pressure." Presented at the 2018 International Space Station (ISS) Research and Development Conference, San Francisco, CA, July 23-26, 2018. 2018 International Space Station (ISS) Research and Development Conference, San Francisco, CA, July 23-26, 2018. , Jul-2018
Abstracts for Journals and Proceedings	Mao XW. "Acute effect of spaceflight on intraocular pressure." Presented at 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Abstracts for Journals and Proceedings	Mao XW. "Spaceflight and radiation induced microvessel and structural damage in ocular tissue." Presented at International Space Station Microgravity Space Medicine Meeting, Manhattan Beach, CA. March 20, 2018. American Institute of Aeronautics &Astronautics (AIAA). International Space Station Microgravity Space Medicine Meeting, Manhattan Beach, CA. March 20, 2018. , Mar-2018